

Predicting Telemarketing Calls Using Client Characteristics

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Introduction

Banks rely on targeted marketing campaigns to promote long-term financial products, but indiscriminate outreach can be both costly and ineffective. Predictive modeling offers an opportunity to allocate marketing resources more efficiently by identifying clients who are most likely to subscribe to a term deposit. Prior research shows that data-driven approaches substantially improve the success of telemarketing campaigns and reduce wasted outreach, while broader work in marketing analytics highlights the power of predictive methods for customer targeting. Motivated by these findings, our project investigates whether subscription outcomes can be reliably predicted using machine learning, with the broader goal of determining when outreach efforts are worth the investment. Our research question is: *Can a machine learning model reliably predict term-deposit subscription outcomes in a way that improves the efficiency of bank marketing outreach?*

Methods

The dataset for this project comes from the UC Irvine Machine Learning Repository and was originally used in a 2014 study analyzing bank telemarketing performance. Although the repository description lists 16 features, our inspection in Pandas confirmed 20 features and 45,211 observations. The feature space includes demographic variables such as age, education, and marital status; financial information such as account balance and loan status; and marketing contact details including call duration and number of campaign contacts. The target variable indicates whether a client subscribed to a term deposit, with 11.3% responding “yes.”

Data cleaning involved handling missing values labeled as “unknown,” encoding categorical variables, and preparing numerical features for modeling. The full dataset was randomly divided into training and test sets at the start of the project to ensure honest evaluation. We trained four models: penalized logistic regression, support vector machine (SVM), gradient boosting, and a neural network. Each model was built within a preprocessing pipeline that included appropriate scaling and encoding steps to prevent data leakage.

Model selection and hyperparameter tuning were carried out using stratified k-fold cross-validation to preserve the class imbalance within each fold. We evaluated models using accuracy, precision, recall, F1-score, and ROC-AUC. After identifying the strongest model, we retrained it on the full training data and evaluated it on the reserved test set.

Results

Across cross-validation, Gradient Boosting emerged as the best-performing model, outperforming logistic regression, SVM, and the neural network. Its superior ROC-AUC, accuracy, and F1-score indicate that models capable of capturing non-linear relationships are particularly effective for predicting subscription outcomes. The table below summarizes the cross-validated performance of each model:

<i>Model</i>	<i>ROC-AUC</i>	<i>Accuracy</i>	<i>F1-Score</i>
Gradient Boosting	0.9470	0.9149	0.5877
Neural Network	0.9413	N/A	N/A
Penalized Logistic Regression	0.9337	0.9104	0.5149
Support Vector Machine	0.9309	0.9042	0.4316

After retraining Gradient Boosting on the full training data, the model achieved strong performance on the untouched test set: accuracy of 0.9229, precision of 0.6915, recall of 0.5700, F1-score of 0.6249, and ROC-AUC of 0.9545. The confusion matrix showed 529 true positives and 7074 true negatives, reflecting the model's ability to correctly classify both subscribing and non-subscribing clients despite the class imbalance.

Discussion

The findings demonstrate that machine learning can successfully predict term-deposit subscription outcomes, with Gradient Boosting offering the most reliable performance. Its ability to model complex feature interactions likely explains its advantage over linear methods such as logistic regression and margin-based methods like SVM. The neural network performed reasonably well but may require further architecture tuning or feature engineering to match the performance of Gradient Boosting.

Several limitations should be noted. The dataset is imbalanced, which naturally limits recall on the positive class. The data also reflects historical telemarketing practices, and client behavior may shift over time due to economic conditions or changes in communication strategies. Additionally, while Gradient Boosting performs well, its interpretability is limited compared to simpler models, which may complicate adoption in real-world banking contexts where transparency is important. Future work could incorporate feature importance analysis, alternative imbalance-handling methods, and model evaluation on more recent or domain-specific datasets.

References

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